



2025

I23(B)

BOOKLET NO.

209075

Chemistry
Paper – II

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional / Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.



SECTION – A

Q1. Answer the following questions. (10×5=50)

- Q1. (a) Write the structure of [10] annulene and comment on its aromatic character. 10
- Q1. (b) When benzaldehyde undergoes Cannizzaro reaction in the presence of NaOH and isotopically labeled water (H_2^{18}O), oxygen-18 is found only in the benzoic acid and not in benzyl alcohol. Explain mechanistically why this happens? 10
- Q1. (c) Explain the major product when tert-butyl bromide reacts with benzene in the presence of AlCl_3 and light. Discuss both ionic and radical pathway. 10
- Q1. (d) The thermal ring opening of cis-3, 4-dimethyl cyclobutene gives a trans, trans 2, 4 hexadiene, but its photochemical reaction gives a cis, trans isomer. Explain this inversion in terms of orbital symmetry and the FMO approach. 10
- Q1. (e) Explain how tacticity (isotactic, syndiotactic, atactic) influences the crystalline and amorphous nature of polystyrene. Support your answer with structural representations. 10

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- Q2. (a) Explain how isotopic labelling can differentiate between elimination-addition and addition-elimination mechanism in nucleophilic aromatic substitution via benzene intermediate. 20
- Q2. (b) Define Tropolones. How does resonance contribute to their stability? 20
- Q2. (c) How does the reaction of configuration in the hydrolysis of thionyl chloride derived alkyl chlorosulfites support the $\text{S}_{\text{N}}1$? 10

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- Q3. (a) Explain why the rearrangement of cyclohexanone oxime gives ϵ -caprolactam while acetone oxime does not yield a cyclic amide. 20
- Q3. (b) Define a sigmatropic rearrangement and explain the notation [ij]. Using this classify the following as [1, 3] or [3, 3] shifts. 20
- a) Claisen rearrangement
- b) Cope rearrangement
- c) Allylic rearrangement of allyl methyl ether.
- Q3. (c) Why does the Wittig reaction with unstabilized ylides favor Z-alkenes while stabilized ylides favour E-alkenes? Support with mechanistic reasoning. 10
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- Q4. (a) Outline the mechanism of the Favorskii rearrangement of α -haloketones and explain the role of enolate formation in determining the product structure. 20
- Q4. (b) Explain the role of hydrogen bonding in base pairing between purines and pyrimidines. Why does A-T pairing involve two bonds while G-C pairing involves three ? 20
- Q4. (c) The HOMO-LUMO gap in a Diel's-Alder reaction decreases upon substituting electron-donating groups on the dienes and electron withdrawing groups on the dienophile. Explain how supports FMO predictions for reaction rate enhancement. 10
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SECTION – B

- Q5. Answer the following questions. (10×5=50)
- Q5. (a) Predict the products and justify your answer. 10
- a) Cyclohexene + OsO_4 (at.)/NMO $\rightarrow$?
 - b) 1, 2-Ethandiol + $\text{HIO}_4 \rightarrow$?
 - c) 3-Pentanol + CrO_3 in pyridine $\rightarrow$?
 - d) Cyclohexene + $\text{SeO}_2 \rightarrow$?
- Q5. (b) What is the Franck-Condon principle ? How does it explain the vertical nature of electronic transitions in photochemical processes ? 10
- Q5. (c) Derive the expression for rotational energy levels of a rigid diatomic molecule and explain how isotopic substitution affects the rotational spectrum. Illustrate with the example of HCl and DCl. 10
- Q5. (d) Determine the number of normal modes of vibration for CO_2 and classify which are IR active. 10
- Q5. (e) Provide a detailed mechanistic pathway for the oxidation of cyclohexanone to ϵ -Caprolactone by MCPBA, indicating the transition state structure and orbital interaction. 10
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- Q6. (a) Compare and contrast the oxidation of vicinal diols by periodic acid (HIO_4) and lead tetra acetate ($\text{Pb}(\text{OAc})_4$) in terms of mechanism, intermediate formation and product type. 20
- Q6. (b) Explain how isotopic substitution (D for H) can be used to confirm NMR assignments of protons in a molecule. Illustrate with an example, showing how chemical shift and splitting pattern change. 20
- Q6. (c) A compound has molecular formula $\text{C}_5\text{H}_{10}\text{O}$ with $M^+ = 86$ IR absorption at 1715 cm^{-1} ($\text{C}=\text{O}$) and ^1H NMR $\delta = 2.1$ ppm (3H, singlet). Deduce the structure of the compound and justify your answer using spectral data. 10
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- Q7. (a) In photochemistry of Butyraldehyde both α -cleavage (Type I) and γ -abstraction (Type II) are possible under what structural and solvent condition does each dominate? 20
- Q7. (b) Using Woodward-Fieser rules calculate λ_{max} for trans-2-hexen-1-one (trans-2-hexen-1-one). 20
- Q7. (c) Calculate the rotational constant β for HCl (bond length = 127 pm) and predict the spacing between adjacent lines in its rotational spectrum. 10
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- Q8. (a) A compound with molecular formula $\text{C}_4\text{H}_{10}\text{O}$, $M^+ = 12$. Show IR absorption at 1735 cm^{-1} and ^1H NMR signals; triplet at 1.2 ppm (3H) and quartet at 2.4 ppm (2H). Propose the structure and rationalize the spectral features. 20
- Q8. (b) Using IR, UV, ^1H NMR and MS data, determine the structure of with molecular formula $\text{C}_8\text{H}_8\text{O}$, $M^+ = 120$, IR absorption at 1720 cm^{-1} , $\text{UV}_{\text{max}} = 245\text{ nm}$ ^1H NMR $\delta = 7.3$ (5H singlet) and $\delta = 2.4$ (3H, singlet). Assign the proton and explain your reasoning. 20
- Q8. (c) Discuss the differences in the reducing behavior of LiAlH_4 and NaBH_4 towards
- esters
 - acid chlorides
 - amides. Provide mechanism at least for one case.
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